浙江大学2016-17秋冬《数据结构基础》期中模拟练习

开始时间	1/1/2016, 12:00:00 AM	**结束时间**	1/18/2038, 12:00:00 AM	**答题时长**	45分钟
考生	–	**得分**	95	**总分**	100

判断题

总分：15　得分：15

1-1 In a binary search tree, the keys on the same level from left to right must be in sorted (non-decreasing) order. (3分)

◉ T　○ F

评测结果：答案正确（3 分）

1-2 For a sequentially stored linear list of length N, the time complexities for query and insertion are $O(1)$ and $O(N)$, respectively. (3分)

◉ T　○ F

评测结果：答案正确（3 分）

1-3 N^2logN and $NlogN^2$ have the same speed of growth. (3分)

○ T　◉ F

评测结果：答案正确（3 分）

1-4 If keys are pushed onto a stack in the order {1, 2, 3, 4, 5}, then it is impossible to obtain the output sequence {3, 4, 1, 2, 5}. (3分)

◉ T　○ F

评测结果：答案正确（3 分）

1-5 In a directed graph, the sum of the in-degrees and out-degrees of all the vertices is twice the total number of edges. (3分)

◉ T　○ F

评测结果：答案正确（3 分）

单选题

总分：65 得分：60

2-1 The result of performing three DeleteMin operations in the min-heap {1,3,2,6,7,5,4,15,14,12,9,10,11,13,8} is: (5分)

- A. 4,5,6,7,8,9,10,11,12,13,14,15
- B. 4,6,5,13,7,10,8,15,14,12,9,11
- C. 4,6,5,12,7,10,8,15,14,9,13,11
- D. 4,5,6,12,7,10,8,15,14,13,9,11

评测结果：答案正确（5 分）

2-2 If an undirected graph G = (V, E) contains 10 vertices. Then to guarantee that G is connected in any cases, there has to be at least __ edges. (6分)

- A. 45
- B. 37
- C. 36
- D. 9

评测结果：答案正确（6 分）

2-3 What is a critical path in an AOE network? (5分)

- A. the shortest circuit
- B. the longest circuit
- C. the shortest path from the first to the last event
- D. the longest path from the first to the last event

评测结果：答案正确（5 分）

2-4 Given a quadtree(四叉树) with 4 nodes of degree 2, 4 nodes of degree 3, 3 nodes of degree 4. The number of leaf nodes in this tree is __. (5分)

- A. 22
- B. 21
- C. 20
- D. 12

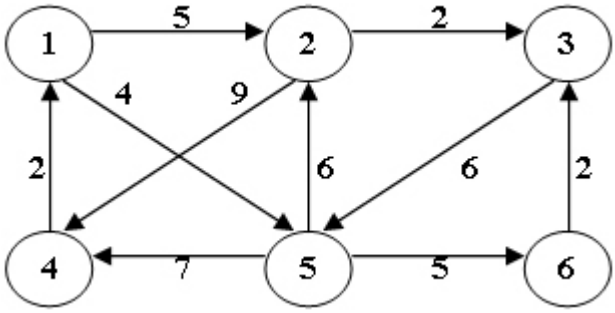
评测结果：答案正确（5 分）

2-5 To insert `s` after `p` in a doubly linked circular list, we must do: (6分)

A. `p->next=s; s->prior=p; p->next->prior=s ; s->next=p->next;`
B. `p->next->prior=s; p->next=s; s->prior=p; s->next=p->next;`
C. `s->prior=p; s->next=p->next; p->next=s; p->next->prior=s;`
D. `s->prior=p; s->next=p->next; p->next->prior=s; p->next=s;`

评测结果：答案正确（6 分）

2-6 Use Dijkstra algorithm to find the shortest paths from 1 to every other vertices. In which order that the destinations must be obtained? (5分)



A. 5, 2, 3, 4, 6
B. 5, 2, 3, 6, 4
C. 5, 2, 4, 3, 6
D. 5, 2, 6, 3, 4

评测结果：答案正确（5 分）

2-7 In a weighted graph, if the length of the shortest path from `b` to `a` is 10, and there exists an edge of weight 3 between `c` and `b`, then how many of the following statements is/are TRUE? (6分)

1. The length of the shortest path from `c` to `a` must be 13.
2. The length of the shortest path from `c` to `a` must be 7.
3. The length of the shortest path from `c` to `a` must be no greater than 13.
4. The length of the shortest path from `c` to `a` must be no less than 7.

A. 1
B. 2
C. 3
D. 4

评测结果：答案正确（6 分）

2-8 Suppose that an array of size `m` is used to store a circular queue. If the head pointer `front` and the current size variable `size` are used to represent the range of the queue instead of `front` and `rear`, then the maximum capacity of this queue can be: (5分)

A. `m-1`

- B. `m`
- C. `m+1`
- D. cannot be determined

评测结果：答案错误（0 分）

2-9 The array representation of a disjoint set is given by { 4, 6, 5, 2, -3, -4, 3 }. If the elements are numbered from 1 to 7, the resulting array after invoking `Union(Find(7),Find(1))` with union-by-size and path-compression is: (5分)

- A. { 4, 6, 5, 2, 6, -7, 3 }
- B. { 4, 6, 5, 2, -7, 5, 3 }
- C. { 6, 6, 5, 6, -7, 5, 5 }
- D. { 6, 6, 5, 6, 6, -7, 5 }

评测结果：答案正确（5 分）

2-10 Insert { 3, 8, 9, 1, 2, 6 } one by one into an initially empty binary search tree. The post-order traversal sequence of the resulting tree is: (6分)

- A. 2, 1, 3, 6, 9, 8
- B. 1, 2, 8, 6, 9, 3
- C. 2, 1, 6, 9, 8, 3
- D. 1, 2, 3, 6, 9, 8

评测结果：答案正确（6 分）

2-11 Given a directed graph G=(V, E) where `V = {v1, v2, v3, v4, v5, v6}` and `E = {<v1,v2>, <v1,v4>, <v2,v6>, <v3,v1>, <v3,v4>, <v4,v5>, <v5,v2>, <v5,v6>}`. Then the topological order of G is: (5分)

- A. `v3, v1, v4, v5, v2, v6`
- B. `v3, v4, v1, v5, v2, v6`
- C. `v1, v3, v4, v5, v2, v6`
- D. `v1, v4, v3, v5, v2, v6`

评测结果：答案正确（5 分）

2-12 In a complete binary tree with 1102 nodes, there must be __ leaf nodes. (6分)

- A. 79
- B. 551
- C. 1063
- D. cannot be determined

评测结果：答案正确（6 分）

程序填空题

总分：20　得分：20

5-1 The function is to lower the value of the integer key at position `P` by a positive amount `D` in a min-heap `H`.

```
void DecreaseKey( int P, int D, PriorityQueue H )
{
   int i, key;
   key = H->Elements[P] - D;
   for ( i = P                                  (5分); H->Elements[i/2] > key; i/=2
      H->Elements[i]=H->Elements[i/2]           (5分);
   H->Elements[i] = key;
}
```

评测结果：答案正确（10 分）

序号	结果	得分
0	答案正确	5
1	答案正确	5

5-2 Please fill in the blanks in the program which performs `Find` as a Union/Find operation with path compression.

```
SetType Find ( ElementType X, DisjSet S )
{
   ElementType root, trail, lead;

   for ( root = X; S[root] > 0; root=S[root]    (5分) ) ;
   for ( trail = X; trail != root; trail = lead ) {
      lead = S[trail] ;
      S[trail]=root                             (5分);
   }
   return root;
}
```

评测结果：答案正确（10 分）

序号	结果	得分
0	答案正确	5

序号	结果	得分
1	答案正确	5